Border Surveillance in India

Current Challenges, Level of Automation, and the Case for AI Video Analytics on Existing CCTV

Background research for the project

IBVAP - Intelligent Border Video Analytics Platform

Smart India Hackathon | Problem Statement ID 26187

AI-Based Intelligent Video Analytics Platform for Border Surveillance using Existing CCTV Infrastructure

Research period covered: 2023 - October 2026

Prepared: October 2026

Sources: 38 cited (government statements, parliamentary answers, defence think tanks, national news, international evaluations)

Contents

1. Executive Summary
2. Research Objectives and Method
3. Current Problems on India's Borders
4. Physical Gaps and Manpower Constraints
5. How Surveillance Works Today: Manual vs Automated
6. Evidence on AI Video Analytics: What It Can and Cannot Do
7. Expected Impact of IBVAP
8. Implications for Our Design
9. Limitations of This Research
10. Conclusion
References

1. Executive Summary

India's border forces face threats that are faster, smaller and more spread out than before, and they still meet them mostly with people. On the Pakistan border, drones carrying drugs and weapons into Punjab rose from 107 recovered in 2023 to about 272 in 2025 [1, 3]. On the Bangladesh border, 864 km is still unfenced and 174.5 km of that cannot be fenced at all because of rivers and terrain [12]. The Myanmar border is about 98% unfenced [11].

Surveillance remains largely manual. The Comprehensive Integrated Border Management System (CIBMS), designed in 2016, has no published coverage figure beyond pilots [17, 19], and the 6,000 km "Smart Border" project announced in May 2026 is still a framework [20]. Confirmed automation exists only in a few places, such as rule-based video alarms at the Attari land port [22].

The strongest evidence for our project is about human limits: operator detection accuracy drops from about 83-84% when watching 4-9 monitors to 64% with 16 [27]. An AI layer on existing cameras changes the operator's job from watching every screen to reviewing flagged alerts. Person and vehicle detection, virtual fences, tracking and number-plate recognition at known hotspots are realistic gains; long-range face recognition on old cameras is the weakest part and should be limited to checkpoints with human review.

2. Research Objectives and Method

We set out to answer four questions before designing IBVAP:

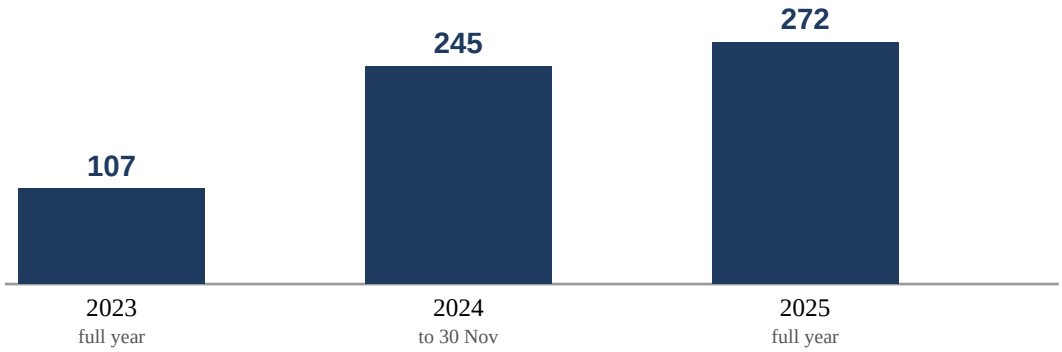
- What security problems are India's border forces facing now, and where?
- Is border surveillance today mostly manual or automated, and where is automation actually deployed?
- What does the evidence say about AI video analytics added to CCTV, in India and abroad?
- Given this, what impact can IBVAP realistically claim, and how should it be designed?

Method. We reviewed government statements and parliamentary answers (Lok Sabha replies, MHA releases), BSF and SSB press briefings, defence think-tank analysis (MP-IDSA), national news reporting, international evaluations of automated border surveillance (US CBP towers, GAO findings), peer-reviewed work on CCTV operator performance, and reporting on face recognition accuracy (Delhi Police, NIST). The period covered is 2023 to October 2026; older data is labelled with its year. Claims that could not be confirmed from a source were left out.

3. Current Problems on India's Borders

3.1 Pakistan border: the drone threat

The Pakistan border (2,289 km, guarded by BSF) now faces an air threat more than a ground one. BSF's Punjab Frontier brought down 107 drones and seized 442 kg of heroin in 2023 [1]. In 2024 it seized 245 drones by 30 November [2]. In 2025 it seized 272 drones, over 367 kg of heroin, more than 10 kg of RDX, 12 grenades and 200 weapons [3]; Western Command counts reached 275-278 for the year [4, 5].



Drones recovered by BSF, Punjab Frontier

Figure 1. Drones recovered on the Punjab border more than doubled between 2023 and 2025. Sources: [1, 2, 3].

The drones are also changing. BSF's Additional Director General says they are getting smaller, cheaper and carry lighter payloads (some around 500 g), use frequency hopping and fly higher, so they are harder to detect; drops now land well inside India [6]. The figures fit this: heroin recovered per drone fell from about 4 kg in 2023 to about 1.3 kg in 2025 (our ratio of the Punjab totals, not an official figure), while arrests of Indian suspects in Punjab, mostly people who pick up the drops, rose from 94 to 251 [4, 5]. In Jammu the main threats are tunnels and riverine infiltration; BSF is planning tunnel-detection sensors and "AI-based night cameras" there [4].

3.2 Bangladesh, Nepal and Myanmar borders

The Bangladesh border is a ground problem. BSF reported more than 10,000 Bangladeshis held while trying to cross illegally (released December 2025, period not stated) [7]. In 2024 BSF seized 172 units of gold and 178 kg of silver on this border [8], and cattle smugglers have attacked BSF personnel in West Bengal [9]. On the open Nepal and Bhutan borders, SSB recorded 1,250 human-trafficking cases, 2,156 victims rescued and 1,094 traffickers arrested between 2020 and 2025 [10]. On the Myanmar border, about 78,731 refugees have entered India since the 2021 coup, and in May 2025 Assam Rifles clashed with armed fighters disrupting fencing work [11].

Border (force)	Main recent problem	Key hotspots	Physical gap
Pakistan (BSF)	Drone drops of heroin, arms and RDX; tunnels; flood-time infiltration	Punjab drone hotspots (Amritsar region); Jammu, Samba, Kathua riverine belts	Little unfenced, but floods breached the fence in 2025 [14]
Bangladesh (BSF)	Illegal crossings; gold, silver and cattle smuggling; attacks on personnel	West Bengal, Tripura, Meghalaya, Assam riverine stretches	864 km unfenced, 174.5 km not feasible to fence [12]
Nepal / Bhutan (SSB)	Human trafficking, illegal crossings	Open border by treaty	Unfenced by design
Myanmar (Assam Rifles)	Refugees, drugs, insurgents	Mizoram, Manipur (Moreh)	About 30 of 1,643 km fenced [11]

Table 1. Summary of problems by border.

4. Physical Gaps and Manpower Constraints

Gaps come mostly from geography and recur in the same places each year. A Lok Sabha answer in February 2025 put the Bangladesh border at 4,096.7 km, of which 864.5 km is unfenced and 174.5 km is "non-feasible" [12, 13]. After August 2024, BSF mapped 635 vulnerable patches covering 484 km across the Pakistan and Bangladesh borders, planning infrared sensors in the east and a new fence design with CCTV, PTZ and bullet cameras in the west [8]. Floods in September 2025 breached the fence in Jammu, Samba and Kathua and cut off border posts; BSF fell back to rear positions with boats and night-vision devices [14]. Winter fog leads to an intensified security grid every December [5].

Manpower is the other constraint. BSF has over 276,000 personnel in 193 battalions covering 6,386 km [5]. Its Western Command ADG admitted in December 2025 that some gaps "might be possible" to exploit because of manpower limits [4]. Between 2018 and 2022, 23,553 BSF personnel left through resignation or retirement and 174 died by suicide [15], and a 2016 parliamentary committee found personnel working 16-18 hour duties [16]. These figures are older, but they describe a force that watches the border through tired people.

5. How Surveillance Works Today: Manual vs Automated

Surveillance on India's land borders is still mostly manual: fences, floodlights, border outposts, foot patrols, ambush positions and handheld thermal or night-vision devices. During Operation Sindoor in May 2025, BSF credited handheld thermal imagers and troops with detecting an infiltration attempt, not automated alerts.

System	Status	Where	Level of automation
Fence, floodlights, patrols, nakas	In use everywhere	All borders	Manual
CIBMS (sensors, cameras, radar, command centre)	Designed 2016; pilots in Jammu; smart fence at Dhubri launched 2019; no published coverage since [17, 18, 19]	Jammu; Dhubri (Assam)	Partial, pilot scale
Rule-based CCTV alarms (wall-scaling, fake plates, unattended objects, crowds)	Deployed; 200+ HD cameras, Rs 9 crore [22]	Attari land port	Automated alerts, no face recognition reported
Anti-drone systems	Being deployed "on a large scale"; unit count not public [6, 23]	Punjab, Jammu	Automated, extent unknown
Drone forensics lab	Operational [6]	Amritsar	Analysis after recovery
"Smart Border" grid (drone radar, thermal cameras, AI surveillance)	Announced May 2026; framework stage; no contracts found [20, 21]	~6,000 km, Pakistan and Bangladesh borders	Planned
AI night cameras, tunnel sensors	Planned (Dec 2025) [4]	Jammu	Planned

Table 2. Current state of border surveillance technology.

A 2017 MP-IDSA brief listed the problems that held back CIBMS: high cost, scarce spare parts, unreliable power, harsh terrain and false alarms in comparable US systems [17]; a 2025 explainer repeated many of them, adding heat in Rajasthan, fog in Jammu and Kashmir and weak coordination between agencies [19]. Even in drone work, BSF's own descriptions stress soldiers watching drop zones and joint road checks with police [24]. Public counts do not separate drones stopped by technology from those found by patrols. The government push for indigenous technology, such as the BSF-IIT Jodhpur MoU [25] and the MyGov border-management challenge [26], is still at the research stage. This is the gap IBVAP is designed for.

6. Evidence on AI Video Analytics: What It Can and Cannot Do

6.1 The human limit

The strongest argument for AI analytics concerns people. In studies, operator detection accuracy falls from 83-84% with 4-9 monitors to 64% with 16. A UK Home Office review found individual operators responsible for 27 to 520 cameras, and in one police control room operators spent over 90% of their time on a single monitor, glancing at the rest for one to four seconds [27]. The often-repeated claim that attention "collapses after 20 minutes" comes from a 1999 US guidance document, not a controlled study [28], so we do not rely on it.

6.2 International and Indian deployments

The closest comparison is the US. Customs and Border Protection's autonomous surveillance towers detect people, vehicles and animals and alert agents; they became a programme of record with 60 towers in 2020 [29], and about $1 billion is planned to expand them through 2034 [30]. Yet no detection rate, false-alarm rate or manpower saving has been published [31], and in 2024 about 30% of cameras on primary towers were inoperable [32]. The lesson: maintenance, uptime and measurement fail more often than algorithms. In India, Andhra Pradesh's CCTV360 network (~15,000 cameras) reports individual successes, such as a stolen vehicle traced in 3 hours and a wanted person located 81 minutes after a face match, but no aggregate accuracy figures [33].

6.3 Face recognition needs caution

Delhi Police told the Delhi High Court its face recognition system was about 2% accurate in 2018 [34], and it treats matches above 80% similarity as leads [35]. NIST's test of 189 algorithms found false-positive rates that vary substantially by race and sex [36]. At night, at long range, in fog and on low-resolution cameras, performance will be well below lab conditions. Section 17 of the DPDP Act 2023 lets the government exempt security agencies [37], so safeguards must be designed in rather than assumed; US border towers failed all six of GAO's privacy protections [38].

7. Expected Impact of IBVAP

Matching each IBVAP capability to a documented problem gives a realistic picture of impact:

IBVAP capability	Problem it addresses	Best-fit locations	Realistic impact
Person/vehicle detection + virtual fence	Crossings through unfenced, riverine and flood-breached stretches; too few people watching screens	174.5 km non-feasible Bangladesh stretch; Jammu riverine belt; 635 mapped vulnerable patches	**High.** Turns cameras that exist but go unwatched into alerts. Depends on coverage, IR/thermal input and power.
Multi-object tracking	Drop pickups inland; smugglers moving away from the border	Punjab drone hotspots and the village roads behind them	**Medium-high.** Targets the people on the ground (251 arrests in 2025) rather than the drone.
ANPR	Vehicles used for pickups and smuggling; fake plates	Attari, Petrapole and other land ports; roads leading from hotspots	**Medium-high.** Already proven as a rule at Attari; adds watchlists and movement history.
Real-time alerts + command dashboard	Slow response; few personnel over long distances	BOP and battalion control rooms	**Medium.** Value depends on response capacity; detection does not guarantee action.
Face recognition	Trafficking, repeat crossers, watchlists	Controlled checkpoints only (ICPs, crossing points)	**Low at open border, moderate at checkpoints** with human review.
Camera-based drone detection	Small, high-flying drones	Punjab	**Low on its own.** Better used to spot the drop and pickup.

Table 3. Expected impact of each IBVAP capability, based on the evidence in sections 3-6.

8. Implications for Our Design

The research directly shaped the following design decisions in IBVAP:

- **Alert review, not screen watching.** The dashboard shows prioritised alerts with snapshots, so one operator can cover many cameras (section 6.1).
- **Incident grouping and false-alarm controls.** Repeated alerts collapse into one incident; animals are ignored; alerts need several confirming frames and a minimum dwell time (CIBMS false-alarm history, section 5).
- **Edge deployment and offline mode.** Processing at the border outpost, with alerts queued when the link drops, because power and connectivity are documented problems (section 5).
- **Night and fog support.** Low-light enhancement and thermal/IR input, because gaps are worst at night and in winter fog (section 4).
- **Face recognition limited to checkpoints**, with a Confirm/Reject step for every match (section 6.3).
- **Accountability by design.** Role-based access, audit log, evidence export with file hashes, and data-retention settings (section 6.3).
- **Measuring its own performance.** The analytics page tracks false-alarm rate, response time and camera uptime, which neither the US towers nor CIBMS have published (section 6.2).
- **Camera health monitoring.** Offline cameras are flagged at once, since about 30% of US tower cameras were found inoperable (section 6.2).

9. Limitations of This Research

- No 2026 statistics were found for any border; trends end with the December 2025 BSF press conferences.
- No official 2024-2026 figure exists in public sources for how many kilometres CIBMS covers.

- Several figures come from news reports of BSF press briefings rather than official annual reports; Myanmar figures rely on one think-tank source [11].
- Staffing, suicide and duty-hour data is older (2016-2022).
- There is no independent cost comparison of retrofitting AI onto existing cameras versus buying smart cameras, and no published measurement of AI performance on legacy Indian cameras at night or in fog.
- For these reasons we do not claim a specific percentage improvement in detection; IBVAP is designed to measure this itself once deployed.

10. Conclusion

The research shows a mismatch. Border threats have become dispersed and frequent: lighter drone drops, inland pickups, crossings through river gaps and flood breaches. Surveillance, meanwhile, is still organised around personnel and a decade-old smart-fence plan that has not moved beyond pilots. The 2026 Smart Border framework shows that the government sees the same need.

IBVAP's value is not a long feature list but being cheap to deploy at known hotspots on cameras that already exist, and showing its own performance. Its realistic promise is that the cameras India already has get watched: detection, virtual fencing, tracking and ANPR in the Jammu riverine belt, the unfenceable Bangladesh stretches, the Punjab drop zones and the land ports, with face recognition kept to checkpoints under human review.

References

All sources accessed October 2026.

[1] Indian Defence News, "Over 100 drones downed, ammunition and drugs seized" (Jan 2024). https://www.indiandefensenews.in/2024/01/over-100-drones-downed-ammunition-drugs.html

[2] All India Radio News, "BSF seizes 245 drones smuggling arms and narcotics from Pakistan in Punjab" (Dec 2024). https://www.newsonair.gov.in/bsf-seizes-245-drones-smuggling-arms-and-narcotics-from-pakistan-in-punjab

[3] All India Radio News, "BSF seizes 272 drones, over 367 kg heroin along Punjab border in 2025". https://www.newsonair.gov.in/bsf-seizes-272-drones-over-367-kg-heroin-along-punjab-border-in-2025

[4] Swarajya, "BSF flags drone hot spots along Punjab border, plans major tech upgrade" (Dec 2025). https://swarajyamag.com/amp/story/news-brief/bsf-flags-drone-hot-spots-along-punjab-border-plans-major-tech-upgrade

[5] The Tribune, "On diamond jubilee, BSF flags drone threat, border crime spike, steps up winter vigil" (Dec 2025). https://www.tribuneindia.com/news/chandigarh/on-diamond-jubilee-bsf-flags-drone-threat-border-crime-spike-steps-up-winter-vigil

[6] The Tribune, "BSF integrating new technologies to strengthen anti-infiltration, anti-smuggling grid in Punjab, Jammu sectors". https://www.tribuneindia.com/news/defence/bsf-integrating-new-technologies-to-strengthen-anti-infiltration-anti-smuggling-grid-in-punjab-jammu-sectors

[7] Deccan Herald / PTI, "Over 10,000 Bangladeshis held while attempting illegal border crossings: BSF" (Dec 2025). https://www.deccanherald.com/india/west-bengal/over-10000-bangladeshis-held-while-attempting-illegal-border-crossings-bsf-3815879

[8] Kashmir Reader, "Plugging gaps on Pak, Bangla borders, BSF launches special surveillance project" (Dec 2024). https://kashmirreader.com/2024/12/07/plugging-gaps-on-pak-bangla-borders-bsf-launches-special-surveillance-project/

[9] The Hitavada, "BSF jawans attacked by cattle smugglers at India-Bangladesh border" (Jan 2025). https://thehitavada.com/Encyc/2025/1/11/BSF-jawans-attacked-by-cattle-smugglers-at-India-Desh-border.html

[10] ETV Bharat, "Union Home Secretary reviews operational preparedness and future strategy of SSB" (May 2025). https://www.etvbharat.com/en/!bharat/union-home-secretary-reviews-operational-preparedness-and-future-strategy-of-ssb-enn25052005762

[11] IMPRI, "The end of the Free Movement Regime: India's Myanmar fencing strategy and its implications (2020-2025)". https://www.impriindia.com/insights/policy-update/the-end-of-the-free-movement-regime-indias-myanmar-fencing-strategy-and-its-implications-2020-2025/

[12] The Sentinel Assam, "Over 864 km of India-Bangladesh border still unfenced: Union Minister Nityanand Rai" (Feb 2025). https://www.sentinelassam.com/amp/story/topheadlines/over-864-km-of-india-bangladesh-border-still-unfenced-union-minister-nityanand-rai

[13] All India Radio News, "Over 3,232 km of India-Bangladesh border fenced, says MoS Home Nityanand Rai" (Feb 2025). https://www.newsonair.gov.in/over-3232-km-of-india-bangladesh-border-fenced-says-mos-for-home-nityanand-rai

[14] Deccan Herald, "Floods damage border fencing in Jammu, BSF on high alert to foil infiltration" (Sep 2025). https://www.deccanherald.com/amp/story/india%2Fjammu-and-kashmir%2Ffloods-damage-border-fencing-in-jammu-bsf-on-high-alert-to-foil-infiltration-3709828

[15] ThePrint, "654 suicides, 50,000 resignations in 5 years: the crisis stalking India's CAPFs" (2023). https://theprint.in/india/654-suicides-50000-resignations-in-5-years-the-crisis-stalking-indias-capfs/1787191/

[16] PRS Legislative Research, report summary "Border Security: Capacity Building and Institutions" (2016). https://prsindia.org/policy/report-summaries/border-security-capacity-building-and-institutions

[17] MP-IDSA Issue Brief, "Comprehensive Integrated Border Management System: Issues and Challenges" (2017). https://idsa.in/publisher/issuebrief/comprehensive-integrated-border-management-system-issues-and-challenges

[18] Ministry of Home Affairs press release on CIBMS (BOLD-QIT), Dhubri, Assam (Mar 2019). https://xn--i1b5bzbybhfo5c8b4bxh.xn--11b7cb3a6a.xn--h2brj9c/sites/default/files/PRESSRELEASEHMCIBMSASSAM_07032019.pdf

[19] Drishti IAS, "India's push for tech-driven border security" (Apr 2025). https://www.drishtiias.com/hindi/daily-updates/daily-news-analysis/india-s-push-for-tech-driven-border-security/print/manual

[20] WION, "India smart border security grid explained" (2026). https://www.wionews.com/india-news/india-smart-border-security-grid-explained-1787765111816

[21] UNI, "India set to launch ISBS as part of Smart Border initiative" (2026). https://test.uniindia.com/india-set-to-launch-isbs-as-part-of-smart-border-initiative/east/news/3866527.html

[22] The Tribune, "No trade for four months, Rs 9 cr CCTVs come up at Attari border" (Aug 2025). https://www.tribuneindia.com/news/punjab/no-trade-for-four-months-rs-9-cr-cctvs-come-up-at-attari-border/amp

[23] All India Radio News, "Punjab launches Baaj Akh anti-drone system to curb cross-border drug smuggling" (2025). https://www.newsonair.gov.in/punjab-launches-baaj-akh-anti-drone-system-to-curb-cross-border-drug-smuggling

[24] The Week, "BSF's war on drones: 255 Pakistani drones neutralised in 2025" (Nov 2025). https://www.theweek.in/news/defence/2025/11/18/bsfs-war-on-drones-255-pakistani-drones-neutralised-in-2025.amp.html

[25] ETV Bharat, "IIT Jodhpur and BSF sign MoU to develop advanced indigenous security technologies" (Nov 2025). https://www.etvbharat.com/en/state/iit-jodhpur-and-bsf-sign-mou-to-develop-advanced-indigenous-security-technologies-enn25110604474

[26] MyGov Innovate India, "Technological Solutions for Border Management" challenge. https://innovateindia.mygov.in/technological-solutions-for-border-management/

[27] Stainer et al., "The development of a cognitive model of CCTV surveillance" (summary), Frontiers in Human Neuroscience (2013). https://www.frontiersin.org/journals/human-neuroscience/articles/10.3389/fnhum.2013.00615/full

[28] NCJRS / OJP, "The Appropriate and Effective Use of Security Technologies in U.S. Schools" (1999). https://ojp.gov/ncjrs/virtual-library/abstracts/appropriate-and-effective-use-security-technologies-us-schools

[29] Defense Daily, "CBP makes autonomous surveillance tower system program of record" (2020). https://defensedaily.com/cbp-makes-autonomous-surveillance-tower-system-program-record/homeland-security

[30] Reason, "Homeland Security is spending $1 billion on autonomous border surveillance towers" (Jul 2026). https://reason.com/2026/07/30/homeland-security-is-spending-1-billion-on-autonomous-border-surveillance-towers/

[31] Axios, "Border Patrol AI on the US-Mexico border" (Dec 2023). https://www.axios.com/2023/12/12/border-patrol-ai-us-mexico-wall-surveillance-virtual

[32] TechRadar (reporting NBC News), "US border surveillance towers face significant operational failures" (Dec 2024). https://www.techradar.com/pro/US-border-surveillance-towers-face-significant-operational-failures

[33] Deccan Chronicle, "AI-powered CCTV360 network strengthens policing in AP" (Jul 2026). https://www.deccanchronicle.com/southern-states/andhra-pradesh/ai-powered-cctv360-network-strengthens-policing-in-ap-1968162

[34] Internet Freedom Foundation, "From investigation to conviction: how does the police use FRT?". https://internetfreedom.in/from-investigation-to-conviction-how-does-the-police-use-frt/

[35] MediaNama, "Delhi Police: 80% accuracy threshold for facial recognition (RTI)" (Aug 2022). https://www.medianama.com/2022/08/223-delhi-police-iff-rti-facial-recognition-80-percent-accuracy/

[36] ASIS Security Management, "Facial recognition error rates vary by demographic" (May 2020, on NIST FRVT Part 3). https://www.asisonline.org/security-management-magazine/articles/2020/05/facial-recognition-error-rates-vary-by-demographic/

[37] MediaNama, "DPDP Bill 2023: government exemptions" (Aug 2023). https://www.medianama.com/2023/08/223-dpdp-bill-2023-government-exemptions/

[38] Electronic Frontier Foundation, "Customs and Border Protection fails baseline privacy requirements for surveillance" (Dec 2024). https://www.eff.org/deeplinks/2024/12/customs-border-protection-fails-baseline-privacy-requirements-surveillance